

ASAHIKAWA, Japan

WMO: 474070

Lat: 43.7569N

Lon: 142.3722E

Elev: 140

StdP: 99.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.5	(c) -15.2	(d) -19.5	(e) 0.7	(f) -17.0	(g) -17.5	(h) 0.8	(i) -14.6	(j) 10.2	(k) -3.3	(l) 8.8	(m) -4.2	(n) 1.1	(o) 80	(p) 0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.6	(c) 29.9	(d) 22.7	(e) 28.3	(f) 21.4	(g) 26.9	(h) 20.6	(i) 23.9	(j) 28.1	(k) 22.9	(l) 26.7	(m) 22.0	(n) 25.3	(o) 3.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	17.6	26.3	21.7	16.6	25.3	20.8	15.7	24.5	72.6	28.2	68.9	27.0	65.4	25.4	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	7.7	6.6	DB	-21.3	32.6	2.2	1.5	-22.8	33.6	-24.1	34.5	-25.3	35.3	-26.9	36.4
			WB	-21.5	24.9	2.1	1.2	-23.0	25.8	-24.3	26.5	-25.5	27.1	-27.1	27.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-7.5	-6.4	-1.6	5.5	12.5	17.2	21.3	21.7	16.8	9.6	2.5	-4.4		
	DBStd	10.88	3.77	3.84	3.65	3.84	3.70	3.33	2.86	2.93	3.33	3.28	3.95	3.55		
	HDD10.0	2249	543	460	361	146	18	1	0	0	1	47	228	446		
	HDD18.3	4276	802	693	619	386	183	59	8	7	68	269	477	704		
	CDD10.0	1274	0	0	0	10	96	217	351	362	203	36	1	0		
	CDD18.3	259	0	0	0	0	2	24	101	110	22	0	0	0		
	CDH23.3	1624	0	0	0	2	56	194	598	663	110	1	0	0		
	CDH26.7	358	0	0	0	0	9	31	135	167	16	0	0	0		
(14) Wind	WSAvg	2.5	2.3	2.5	2.9	2.9	2.9	2.5	2.3	2.2	2.2	2.4	2.8	2.6		
(15) Precipitation	PrecAvg	1092	71	56	55	53	66	72	111	153	135	108	116	98		
	PrecMax	1436	134	115	113	119	134	133	348	478	298	213	226	164		
	PrecMin	731	21	19	16	11	12	11	22	22	26	38	65	47		
	PrecStd	164	25	24	25	23	25	28	62	91	68	37	32	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	5.5	11.2	22.4	27.7	29.3	31.6	32.2	28.7	21.2	15.9	6.7		
		MCWB	1.5	2.4	5.9	12.0	16.4	19.0	23.6	24.3	22.5	15.1	11.1	4.7		
	2%	DB	0.9	3.1	8.2	18.8	24.7	27.3	29.7	30.1	26.2	18.8	12.8	4.2		
		MCWB	-0.5	0.8	3.7	10.8	14.9	18.7	22.7	23.3	20.1	13.8	9.3	2.5		
	5%	DB	-0.5	1.4	6.1	15.4	22.4	25.6	28.1	28.5	24.3	17.1	10.7	2.4		
		MCWB	-1.9	-0.5	2.7	9.0	13.9	17.9	21.6	22.4	19.1	12.6	7.9	0.9		
	10%	DB	-1.9	-0.2	4.3	12.7	19.9	23.8	26.7	27.0	22.5	15.4	8.7	0.9		
		MCWB	-3.1	-1.8	1.5	7.3	12.8	17.2	21.0	21.4	18.0	11.9	6.5	-0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	3.0	6.4	13.5	17.6	21.2	25.0	25.2	23.0	16.4	12.0	4.7		
		MCDB	3.2	5.0	10.4	20.3	25.2	26.5	29.7	30.5	27.4	19.4	14.9	6.2		
	2%	WB	-0.3	0.9	4.4	11.3	15.9	19.8	23.7	24.2	21.6	14.7	9.7	2.7		
		MCDB	0.8	2.7	7.3	17.2	22.5	25.0	27.7	28.7	24.8	17.5	12.3	3.9		
	5%	WB	-1.8	-0.4	2.9	9.5	14.7	18.8	22.8	23.3	20.2	13.5	8.1	1.0		
		MCDB	-0.7	1.2	5.6	15.2	20.6	24.0	26.5	27.2	22.7	16.2	10.4	2.1		
	10%	WB	-3.0	-1.7	1.7	7.6	13.6	17.8	21.9	22.4	18.8	12.2	6.5	-0.3		
		MCDB	-2.0	-0.2	4.1	12.1	18.8	22.6	25.2	25.8	21.7	15.0	8.5	0.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.6	8.4	9.9	11.3	10.2	9.0	8.6	9.4	9.4	6.8	6.3		
		MCDBR	8.0	9.5	10.7	15.5	15.9	13.6	11.0	10.5	11.0	11.8	10.9	8.0		
	5% WB	MCWBR	6.9	7.8	7.2	8.3	7.5	6.3	5.0	4.7	5.6	7.1	7.8	6.7		
		MCWBR	7.8	9.3	9.8	14.4	13.5	11.4	9.0	9.1	9.4	10.2	10.4	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.387	0.452	0.510	0.524	0.510	0.492	0.461	0.406	0.401	0.380	0.346		
		taud	1.968	1.817	1.728	1.781	1.870	1.991	2.108	2.212	2.315	2.181	2.108	1.982		
	Ebn at Noon		742	758	761	758	763	776	785	797	816	754	689	677		
		Edh at Noon	116	162	203	208	196	175	154	134	112	112	102	104		
(44) All-Sky Solar Radiation	RadAvg		1.35	2.19	3.40	4.45	4.96	5.03	4.68	4.29	3.54	2.44	1.36	1.07		
	RadStd		0.15	0.25	0.33	0.41	0.36	0.50	0.55	0.28	0.27	0.17	0.14	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page